unverifiable ideas about what was actually in the world—it simply described the world.

Einstein had read Mach's *History of Mechanics* as a student and was deeply impressed with his criticism of the Newtonian ideas of absolute space and time. "This book exercised a profound influence on me," he wrote decades later. Taking Mach's ideas about eliminating extraneous unobservable entities to heart, Einstein had tackled the problem of the aether, finding it to be an unnecessary hypothesis in special relativity. And, better still, special relativity also consigned to oblivion the absolute space and time that Mach had so despised.

Einstein had, in short, used Mach's ideas to brilliant effect. Machians took inspiration from his work for years, thinking that relativity's success vindicated their approach to the world. Mach's views, they figured, were obviously shared by Einstein, since they played such an important role in his most famous and profound work. But when Mach's followers actually spoke with Einstein himself, they were surprised to find that he was not a dogmatic Machian after all—far from it. Although his theory of relativity dismissed the idea of absolute space and time, it replaced those notions with a different absolute: *spacetime*, a combination of space and time, which is the same for all observers. And the name "relativity" itself, which suggests a rejection of absolutes, was introduced by the physicist Max Planck, not Einstein—Einstein disliked the name "relativity" precisely because it connoted a kind of relativism. He preferred the name "invariant theory," which conjures up a very different set of associations. (The "invariants" in relativity are quantities like spacetime that all observers agree upon—and there are many of these in the theory.) And, later in life, Einstein himself said repeatedly that he did not think that Mach's ideas were to be taken too seriously. "Mach's epistemology… appears to me to be essentially untenable," Einstein wrote. "It cannot give birth to anything living. It can only exterminate harmful vermin." While Mach believed that physics was merely about organizing perceptions of the world, to Einstein, physics was about the world itself. "Science," he said, "has the sole purpose of determining what *is*."

But perhaps most convincing and revealing about Einstein's true stance toward Mach in 1905 are his two other celebrated papers from that year. In one, Einstein explained Brownian motion, the random motion of